

C21EM Econometrics Project – Gender Pay Gap

Samir Khalafalla H00371793

Word count: 4504

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1. Introduction:

The gender wage gap is a major labour economic issue. Gender wage inequality remains one of the most widely studied issues in labour economics. The women's labour market has improved significantly over time, however there's still wage gap between men and women. The wage gap is often used as a measure of gender inequality in labour markets. Differences in education and experience used to explain much of the gap. Today a portion of that gap remains unexplained.

A huge amount of empirical research has been sought to identify the source of the gender wage gap. Some early contributions focus more on differences in human capital, specifically gaps in education, labour market experience and occupational choice which has been the primary explanations for observed wage disparities. These endowment gaps have been narrowed considerably in recent decades. There's been an increasing share of the wage differential which has remained unexplained by observable characteristics.

This project plans to investigate the gender wage gap using the Wooldridge "wage1" dataset, a sample of 526 workers in the United States. Employing OLS regression and the Blinder-Oaxaca decomposition (Blinder, 1973; Oaxaca, 1973), the analysis proceeds in several steps. The project starts with a raw wage gap estimated between male and female workers. This is followed by human capital controls from education, experience and tenure. There are introduced to isolate the adjusted penalty. Thirdly, the Oaxaca-Blinder decomposition separates the total gap into an endowment effect, due to differences in observable characteristics, and a coefficient effect. Showing differential returns which may capture discrimination or unmeasured structural factors. These results contribute to the broader debate on whether the persistence of the gender wage gap is primarily a productivity narrative or a structural and institutional one.

The project's structure is as followed. Section 2 reviews the economic literature on the gender wage gap covering human capital, structural and institutional explanations. Section 3 focuses on the economic and the econometric framework used in this project. This includes the Mincer wage equation, the OLS specifications and the Blinder-Oaxaca decomposition. Section 4 describes the data and presents descriptive statistics. Section 5 reports on the estimation results and section 6 summarises the report as a whole and discusses future research.

2. Literature review:

The gender wage gap is still a huge topic despite the merging in education and labour force participation since the 1980s, a significant difference in earnings persists, which is highlighted more at the top of the wage distribution (Blau and Kahn, 2017). The literature explaining this follows three major approaches: experience and job differences; structural explanations focusing on nonlinear pay and segregation; and more recent institutional analyses of stakeholder monitoring and discrimination. All these points offer genuine insight, but none fully accounts for the lingering puzzle remains after controlling for observable characteristics. This review evaluates these contradictory explanations, and which is motivated by Oaxaca-Blinder decomposition to quantify what observable differences can and cannot explain.

Long-run evidence shows that there has been both substantial merging and persistent inequality. Using PSID and CPS data, Francine D. Blau and Lawrence M. Kahn (2017) document that the female-to-male earnings ratio increased from roughly 62-64% in 1980 to approximately 79-82% by 2010. They also show that convergence was uneven across the distribution, by 2010 the wage gap was noticeably larger at the top than at the median. This could be due to a snowball effect from earlier decades. It raises the question that If educational convergence has largely occurred, why does the gap remain particularly pronounced among high earners?

Mincer and Polachek (1974) explain the gender wage gap by using the differences in expected lifetime labour force attachment and human capital investment. Later evidence does suggest that his explanation has become less sufficient over time since Blau and Kahn (2017) show that although experience differences once explained a substantial part of the gap, their explanatory power had fallen quite sharp by 2010, this implies that human capital theory remains important historically but isn't as impactful for in contemporary wage inequality.

Blinder and Oaxaca (1973) present the core decomposition framework used to separate wage differentials into explained and unexplained components. Later the work raises eyebrows about how literally the unexplained component should be interpreted. Meara, Pastore and Webster (2020) argue that standard decompositions can be overstated the unexplained gap when men and women are not directly comparable in observed characteristics, this suggests that the decomposition is useful as an accounting exercise but less definitive as a measure of discrimination.

Another methodological limitation is the index number problem used in the Oaxaca-Blinder decomposition. The size of the explained and unexplained components depends on the choice of reference wage structure. It can be determined through whether male female or pooled coefficients are treated as the non-discriminatory benchmark. Different choices can yield different decompositions. As originally noted by

Blinder (1973) and Oaxaca (1973) this uncertainty complicates interpretation of the coefficient effect as a pure measure of discrimination. So, while decomposition is a useful way to break down the wage gap, we should be careful about drawing strong conclusions from it.

Meara, Pastore and Webster (2020) argue that traditional decomposition methods exaggerate the unexplained part because men and women in the data are not always directly comparable, they suggest using matching methods instead. Their findings do propose the idea of a single uniform gender pay gap is too simple, as wage disparities interact with a lot more factors other than pay, like part time work, union membership and occupational segregation.

Goldin's (2014) argument is that the remaining gender wage gap is driven by the difference in job structures, which are jobs that disproportionately reward long inflexible hours in which she terms "greedy work". In occupations where earnings are nonlinear in hours worked, people who supply continuous and extended hours receive disproportionately high compensation, this results in disadvantaging workers with greater need for progressive flexibility. This structure diverts our attention away from the individual human capital deficits and more toward the institutional features of wage determination. Blau and Kahn (2017) do further emphasise the importance of occupational and industrial segregation, this shows that sorting across sectors continues to explain a meaningful share of this wage gap. Experimental and audit studies on top of this suggest that discrimination remains a highly relevant mechanism. Which is highlighted in hiring and promotion decisions, which reinforces wage inequality beyond observable productivity differences.

Maoret et al. (2024) show how greater financial analyst coverage reduces the executive gender pay gap by increasing stakeholder attention and reducing information asymmetries in executive labour markets. This indicates that externally monitoring mechanisms may constrain discriminatory pay practices higher up in the firms. Dias, Joyce and Parodi (2020) show that in the UK the gap widens after the first child and work experience explains a substantial share of this divergence. The economic consequences of the gender pay gap have extended beyond individual earnings to broader welfare outcomes. What we can see is that each framework builds on the last rather than replacing it, while also seeing its limits. Seeing all these factors like human capital, occupational sorting and institutional structure suggest that the gap isn't a measurement failure but deeply rooted in labour market inequalities. You could even argue it's culture that observable data alone may never fully capture.

Amaro et al. (2024) show that eliminating the pay gap would increase female earnings and reduce poverty rates by a substantial amount, which highlights that the macroeconomic and distributional implications of wage inequality.

Using microdata across 134 countries, Henrik Kleven et al. show that men and women typically diverge sharply in labour-market outcomes after parenthood, with large cross-country variation in magnitude, implying that family-related constraints can be a dominant driver of gender inequality in many settings.

When looking at international evidence, the gender pay gap depends on institutions, sector and family related labour supply. Chi and Li (2014) talk about China, where the gap has widened and it's underestimated when female employment selection is ignored. Pavlenkova, Alfieri and Masso (2024) provide further evidence that firm-level factors matter, showing that investments in automation in Estonia are associated with a widening of the gender pay gap, particularly in managerial positions. Zimmermann (2022) suggests that female managers can actually help narrow wage gaps within firms. Looking at this trend if jobs that require more manual labour had managerial female workers this would narrow the gap, but would it fix it? Tansel, Keskin and Ozdemir (2020) argue that the gender wage patterns differ across sectors and quantiles in Egypt. All these papers show that the gap is there and it's not dependent on the country but might be an underlying issue of less women in management.

All in all, the literature reveals a progression from productivity-based explanations toward structural and institutional accounts. But above all else, a substantial unexplained residual persists. This could reflect discrimination, unmeasured productivity differences, or structural wage setting mechanisms remains contested. This mystery motivates the empirical strategy used here. Applying the Oaxaca-Blinder decomposition to quantify the relative contribution of observable characteristics and differential returns in explaining the gender wage gap.

3. Model and Framework

The main economic question is whether there is a systematic penalty that remains on female workers wages after accounting for differences. In terms of productivity-related characteristics, If the gender wage gap could be explained by differences in education, experience, and tenure, this means the labour market rewards men and women equally for their skills. If a substantial gap remains even after controlling these characteristics, then it implies that either unmeasured factor, workplace pay structures or discriminatory practices are at work. The framework that has been adopted here uses OLS regression and the Blinder-Oaxaca decomposition since my lecturer recommended this approach and found papers that applied it to a similar question.

3.1 Economic Framework

Standard labour economics explains the wage differences between workers by looking at human capital, it accounts for their skills and knowledge they've built up overtime through education and work. Workers with more years of schooling are expected to earn more because education is acquiring knowledge in turn raises productivity and critical thinking. Workers with more labour market experience are expected to earn more because they learn on the job making them more productive. It is the same with tenure, where an employer may raise wages to reflect firm-specific human capital, as workers acquire skills that are most valuable within that firm.

When this is applied to the gender wage gap, this framework suggests that if men and women differ systematically in their average levels of education, experience, or tenure. Some portion of the observed earnings may simply reflect the workers productivity rather than unequal treatment. Mincer and Polachek (1974) speak on this argument, saying that anticipated labour market interruptions for family reasons could rationally lead women to invest less in the market-oriented part of human capital, thereby generating a wage gap. But as discussed in the literature review, this argument has weakened as women's educational attainment and labour force attachment has increased significantly.

3.2 The Mincer Wage Equation

The theoretical basis for this analysis is the Mincer earnings function, which models log wages as a function of human capital characteristics. This model equates the logarithm of wages as a linear function of years of schooling and a quadratic function of labour market experience. The log linear specification is standard for two reasons. First, hourly wage distributions are usually skewed to the right, and taking logs reduces this skewness bringing the distribution closer to normality. Secondly, the coefficients on the

regressors are interpreted as approximate percentage effects on wages, which is more intuitive than level effects. The mincer framework provides both the theoretical and practical justification for the empirical specification used here because it directly models the relationship between earnings and the productivity-related characteristics we want to test.

3.3 Econometric Specification

This analysis begins as basic as possible to establish the unconditional wage gap

$$\ln(wage_i) = \beta_0 + \beta_1 female_i + u_i$$

In this equation wage is the natural log of the hourly wage and female is an indicator equal to one for female workers and zero for male workers. The coefficient β_1 shows the average difference in log wages between men and women. because the dependant variable is in log terms and the regressor is in binary, the approximate percentage wage differentiation is given by $100 \times (\exp(\beta_1) - 1)$. Robust standard errors are used throughout because the variation in wages sometimes differs across workers, this means that some groups have more spread in their earnings than others

3.4 Estimating the Raw and Adjusted Gender Wage Gap

For the next specification we isolate the adjusted gender wage penalty, the model is extended by adding controls for education, experience and tenure:

$$\ln(wage_i) = \beta_0 + \beta_1 female_i + \beta_2 educ_i + \beta_3 exper_i + \beta_4 tenure_i + u_i$$

in this specification education measures years in schooling, experience captures the total labour market experience and tenure records years with current employer. With these controls β_1 measures the remaining wage gap between men and women with the same levels of schooling experience and tenure. If β_1 falls it shows that part of the raw gap is attributable to observable human capital differences. If a large penalty is there that means that factors beyond observable human capital differences are contributing to the wage gap.

3.5 Nonlinear Returns to Experience

the mincer framework predicts that experience boosts wages a lot in the early years but the effect gets smaller over time. Wage growth is steep early in careers then it flattens as workers age. To show this the model includes the square of experience:

$$\ln(wage_i) = \beta_0 + \beta_1 female_i + \beta_2 educ_i + \beta_3 exper_i + \beta_4 exper_i^2 + \beta_5 tenure_i + u_i$$

When the experience coefficient is positive and the experience squared coefficient is negative that shows that wages rise with experience but at a slower rate which is what we would expect. Adding this curved relationship also gets us a more accurate estimate

of the wage gap because the model is accounting for natural flattening of earnings growth separate from the effect of gender

3.6 Blinder-Oaxaca Decomposition

The OLS decomposition establishes whether a gap exists between genders and how much if it is explained by observable characteristics. It doesn't formally quantify what share of the gap is attributable to differences in those characteristics as opposed to differences in how the specific characteristics are rewarded. The Blinder-Oaxaca decomposition tackles this by separating the mean wage gap into 3 components:

$$\bar{Y}_m - \bar{Y}_f = (\bar{X}_m - \bar{X}_f)\hat{\beta}_f + \bar{X}_f(\hat{\beta}_m - \hat{\beta}_f) + (\bar{X}_m - \bar{X}_f)(\hat{\beta}_m - \hat{\beta}_f)$$

This decomposition splits the average pay gap to 3 parts. The first term, $(\bar{X}_m - \bar{X}_f)\hat{\beta}_f$, is the endowment effect. It is how much the gap would close if women and men had the same levels of education, experience and tenure.

The second term $\bar{X}_f(\hat{\beta}_m - \hat{\beta}_f)$ is the coefficient effect. This captures how big the gap is because the labour market rewards the same characteristics more generously for men than women, this part of the formula is the closest to discrimination or structural inequality.

The third term $(\bar{X}_m - \bar{X}_f)(\hat{\beta}_m - \hat{\beta}_f)$ describes the part of the gap that exists because men both have more of certain characteristics and are rewarded more for them at the same time.

3.7 Identification Issues and Limitations

There's been limitations that apply to this framework. OLS assumes that the variables included in the model are not systematically related to the factors that have been left out. This assumption is unlikely to hold perfectly because wage1 doesn't contain more controls like occupation, industry, part time status or family characteristics which also may affect wages and differ systematically between men and women. Taking these variables away may upwardly bias the estimated female wage penalty by attributing to gender what is partly driven by occupational sorting or hour's structure.

On top of this, the data is cross sectional. This means that workers are only observed at one point in time. this means that the analysis can't control for unobserved individual heterogeneity such as innate ability, motivation or career preferences, which panel data and fixed effects methods would allow. It also means the wage gap can't be tracked over time. So, it can't take trends or dynamics into account.

Thirdly in the Oaxaca-blinder decomposition, the unexplained component shouldn't be interpreted as a direct measure of discrimination. The way it captures the portion of the wage gap that can't be accounted for by the included regressors which may reflect genuine discrimination unmeasured productivity differences or structural wage setting mechanisms that aren't observable in this dataset isn't direct enough.

Another concern is that the experience variable in wage1 looks like its measured as potential experience rather than time spent in employment. If women are more likely to have had career interruptions that are not captured by this measure, their true labour market experience would be overstated. This would mean that in turn the cause of the adjusted gender wage penalty could be understated.

Future work isolating the contribution of education by estimating education as the sole control, would be nice as it complements the mincer framework used here.

Lastly the decomposition is sensitive to the choice of reference wage structure. This project uses female coefficients as the benchmark in line with Stata default, but different choices do yield different splits between the explained and unexplained components.

4. The data

The data used is from the wage1 dataset from Wooldridge Introductory Econometrics textbook. It's a cross-sectional sample of 526 workers that are based in the United States. The information given is hourly wages, years of education, labour market experience, job tenure and gender. The dataset is well suited to the research question since it contains the key variables required for the Mincer earnings framework and includes a binary gender indicator. This allows for both OLS estimation and Oaxaca-Blinder decomposition. One limitation is that the data does not include controls for occupation, industry or union membership, which restricts the set of observable characteristics available for the analysis.

4.1 Data Section Tables

Table 1: Summary Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
wage	526	5.896	3.693	0.53	24.98
lwage	526	1.623	0.532	-0.635	3.218
educ	526	12.563	2.769	0	18
exper	526	17.017	13.572	1	51
tenure	526	5.105	7.224	0	44
female	526	0.479	0.500	0	1

Notes: wage is hourly wage in dollars; lwage is the natural log of wage; educ is years of education; exper is years of labour market experience; tenure is years with current employer; female is a binary indicator (1 = female, 0 = male).

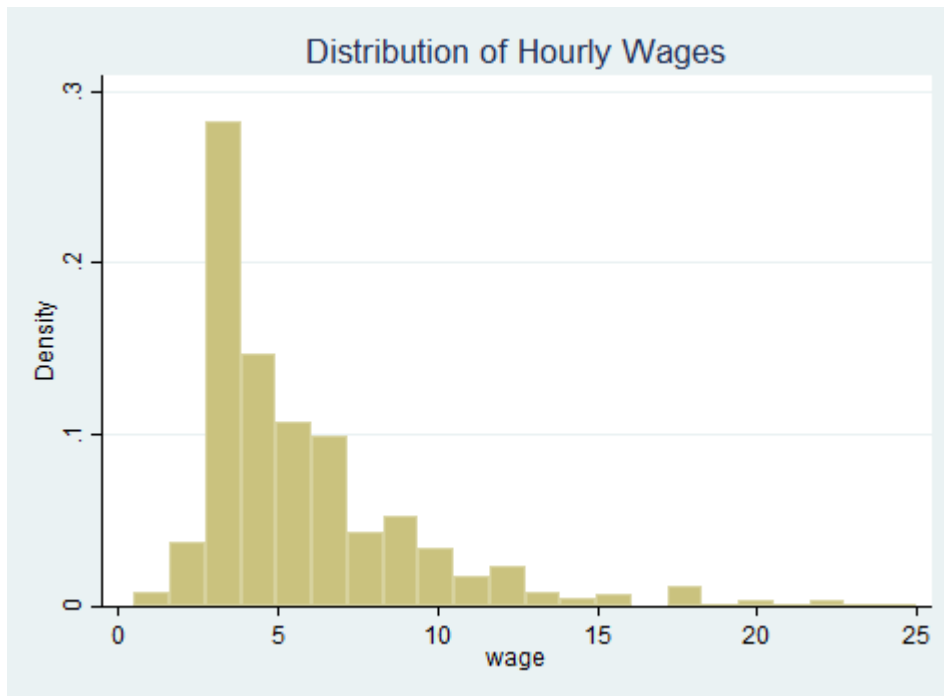


Figure 4.1 shows the distribution of hourly wages.

Table 1 is reporting the summary statistics for the full sample. The distribution of wages ranges from \$0.53 to \$24.98. the average hourly wage is \$5.90 with a standard deviation of \$3.69, with this information we can confirm the positive skewness of 2.01 which shows that there is a wage distribution which is right skewed. This justifies the use of log wages as the dependent variable in the regression analysis. The average worker in the sample has 12.6 years of education, 17 years of market experience and 5.1 years of tenure with the current employer. The sample is roughly evenly split by gender, with women making up around 48% of the observations.

Table 2: Summary Statistics by Gender

	Male	Female	
Variable	Mean	Mean	Difference
lwage	1.814	1.416	0.397
educ	12.788	12.317	0.471
exper	17.558	16.429	1.130
tenure	6.474	3.615	2.859
wage	7.099	4.588	2.511
N	274	252	

Notes: Difference = Male Mean – Female Mean.

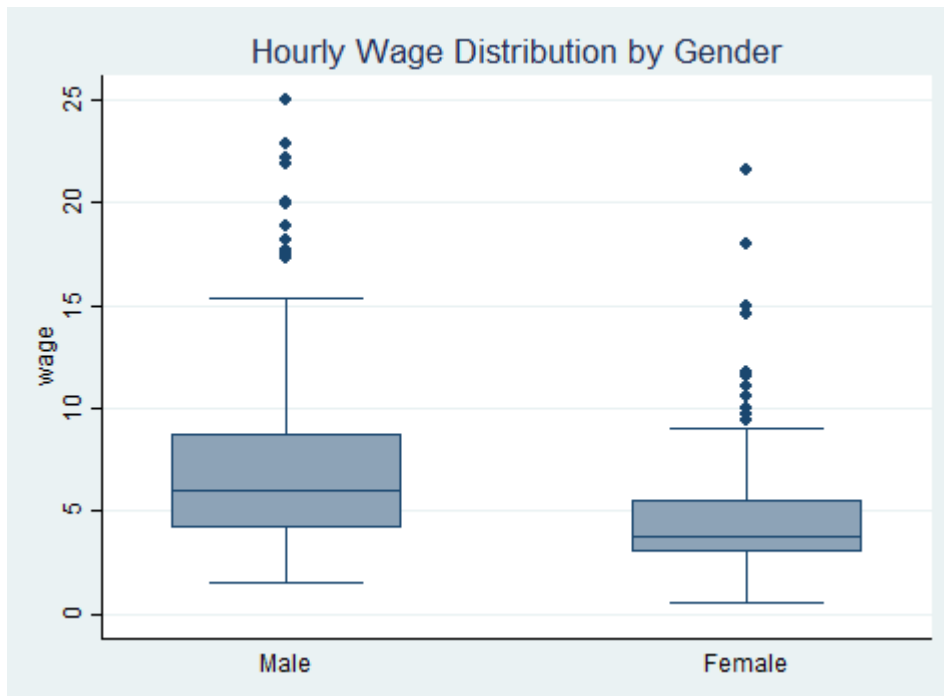


Figure 4.2 shows the right-skewed distribution of hourly wages distributed by gender

Table 2 compares the mean characteristics of male and female workers. The thing that stands out the most is in log wages, where men average 1.814 compared to 1.416 for women, the difference of 0.397 log points. This corresponds to around 33% wage disadvantage for female workers. Men have slightly more education at 12.8 against 12.3 years, marginally more experience at 17.6 to 16.4 years, and notably higher tenure at 6.5 compared to 3.6 years. the largest gap in raw terms is hourly wages, where men earn \$7.10 compared to \$4.59 for women. This difference suggests that part of the wage gap may be attributable to differences in human capital endowments, whether these differences fully account for the gap is an empirical question. The regression analysis in section 5 will formally test this by introducing human capital controls and decomposing the wage differential into explained and unexplained components.

5. Results and interpretation

Table 3: OLS Regression Results - Dependent Variable: $\ln(\text{wage})$

Regressor	Model 1	Model 2	Model 3
female	-0.397 (0.043)	-0.301 (0.038)	-0.298 (0.037)
educ		0.087 (0.008)	0.081 (0.008)
exper		0.005 (0.002)	0.033 (0.005)
exper ²			-0.0006 (0.0001)
tenure		0.017 (0.004)	0.016 (0.003)
Intercept	1.814 (0.032)	0.501 (0.115)	0.415 (0.107)
R ²	0.140	0.392	0.434
RMSE	0.494	0.416	0.402
N	526	526	526

Notes: Heteroskedasticity-robust standard errors in parentheses. Model 1 is the raw gender wage gap with no controls. Model 2 adds human capital controls (education, experience, tenure). Model 3 adds the square of experience to allow for nonlinear returns. All coefficients significant at the 1% level.

Table 3 details the OLS regression results across the three different specifications. Model 1 estimates the raw gender wage gap with no controls. The coefficient on female is -0.397 with a robust standard error of 0.043, which is statistically significant 1% level. This means that the average women earn approximately 33% less than men in hourly wages. An R-squared of 0.140 indicates that gender alone explains about 14% of the variation in log wages. While this confirms a substantial unconditional gap this does tell us that nothing about whether the gap reflects differences in productivity related characteristics or something else.

The second model introduces human capital controls for experience, education and tenure. The female coefficient falls to -0.301 (SE = 0.038), meaning the gap remains significant but it still has shrunk. The reduction from -0.397 to -0.301 suggests that roughly a quarter of the raw gap is attributable to differences in observable human capital. Every year of education is associated with approximately 8.7% higher wages;

this remains consistent with standard human capital predictions. Experience and tenure are both positive and significant but the return to experience is small at 0.5% per year. The R-squared rises substantially to 0.392, indicating the human capital variables explain a considerable share of wage variation. Even though this is improved, a penalty of around 26% for female workers persists even when comparing men and women with the same education, experience and tenure.

The third model adds the square of experience to allow for non-linear returns. With this the experience coefficient rises to 0.033 and the experience squared is -0.0006, both significant at the 1% level. this shows that wages rise with experience, but the gains get smaller over time, which is what we'd expect since early career years tend to matter more for wage growth. The female coefficient barely changes from -0.301 to -0.298, this suggests that the nonlinear specification does not alter the estimated gender penalty in a meaningful way. The R-squared does increase to 0.434, showing an improved model fit. All the coefficients across the three models are estimated with standard errors that are small relative to the coefficient magnitudes and confidence intervals that do not cross zero for any variable.

Theres still a lot of unexplained issues in the female penalty across all three specifications. This still is consistent with Blau and Kahn's (2017) findings that human capital variables now explain only a modest share of the gender wage gap. In this sample, controlling for education, experience and tenure reduces the gap by only roughly one quarter, but a substantial penalty remains. This does motivate the decomposition analysis to formally separate what differences are observable that can and cannot explain.

Table 4: Blinder–Oaxaca Decomposition of the Gender Wage Gap

Component	Coefficient	Std. Err.	z	P > z
Overall				
Male mean (Group 1)	1.814	(0.032)	55.94	0.000
Female mean (Group 2)	1.416	(0.028)	50.37	0.000
Difference	0.397	(0.043)	9.26	0.000
Endowments	0.070	(0.025)	2.73	0.006
Coefficients	0.291	(0.037)	7.79	0.000
Interaction	0.037	(0.022)	1.65	0.099
Endowments Detail				
educ	0.038	(0.020)	1.86	0.062
exper	0.003	(0.004)	0.73	0.465
tenure	0.029	(0.018)	1.65	0.100
Coefficients Detail				
educ	0.200	(0.210)	0.95	0.340
exper	0.096	(0.055)	1.76	0.078
tenure	0.029	(0.027)	1.08	0.281
Intercept	−0.034	(0.233)	−0.15	0.883

Notes: Group 1 = Male (N = 274), Group 2 = Female (N = 252). Decomposition uses female coefficients as the reference wage structure (Stata default). Robust standard errors in parentheses.

The fourth table represents the Blinder-Oaxaca decomposition results. The total mean of the log wage gap between men and women is 0.397, this is statistically significant $z = 9.26$ and a p of 0. This gap was decomposed into three components. The endowments effect is 0.070 which is significant at the 1% level and accounts for around 17.5% of the total gap. This means if women had the same average levels of education experience and tenure as men but were rewarded with female rates the wage gap would shrink by only 0.070 log points. The coefficients effect is 0.291, which is highly significant and accounts for approximately 73% of the total gap. This talks about a portion of the gap which is responsible for men and women receiving different returns for the same characteristics. The interaction term is 0.037 which is marginally significant because it is at the 10% level and it accounts for the remaining 9.3%.

Looking at the endowment detail, education contributes the most at 0.038 followed by tenure at 0.029, while experience contributes very little at 0.003. The tenure result is consistent with Table 2 which showed that men have nearly twice as much tenure as women on average. However, none of the individual endowment components are individually significant at the 5% level, suggesting that while the overall endowment effect is significant, no single characteristic drives it conclusively.

The dominant role of the coefficient's component is the most important finding. It implies that the bulk of the gender wage gap in this sample is not because women have less education or experience than men, but because the labour market appears to reward those same characteristics differently depending on gender. This is consistent with Blau and Kahn's (2017) evidence that the unexplained component has grown as a share of the total gap over time and with Goldin's (2014) argument that structural features of wage determination rather than individual endowment differences are the primary driver of contemporary wage inequality. However as discussed in Section 3.7, this unexplained component should not be interpreted as a direct measure of discrimination. It may also reflect unmeasured productivity differences, occupational sorting, or structural features of the labour market that are not captured in this dataset. Meara, Pastore and Webster (2020) caution that standard decompositions can overstate the unexplained gap when men and women are not directly comparable, which may apply here given the limited set of controls available.

6. Summary

The project has investigated the gender wage gap using the Wooldridge dataset of 526 US workers. This analysis has employed OLS regression across three specifications and the Blinder-Oaxaca decomposition to examine whether the observed wage differential between men and women can be explained by difference in human capital characteristics.

The raw gender wage gap is estimated at 0.397 log points; this corresponds with approximately a 33% wage disadvantage for female workers. This is controlled for education, experience and tenure reduced the gap to around 0.298 log points, this means the observable human capital differences account for roughly a quarter of the raw differential. The remaining penalty of around 26% persists even when comparing men and women with the same level of schooling, experience and tenure. The Oaxaca-Blinder decomposition confirmed this, the pattern shows that the endowment difference explains only 17.5% of the total gap while the coefficient effect accounts for 73%. This suggests that the gap is driven more by differential returns to characteristics than by differences in the characteristics themselves.

In the future addressing these limitations by using richer datasets would enrich the results. Including occupational controls, by employing panel data methods to control for unobserved heterogeneity, or by applying quantile regression to examine whether the gap varies across the wage distribution. A regression including only education as a control would give an independent contribution to the gap, which can be left for future analysis. This analysis demonstrates that a substantial wage penalty persists in this sample that cannot be accounted for by standard human capital variables alone on top of its limitations.

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Appendix: Stata Code and Output

Stata code

```
clear all
```

```
* create path, log and load project
```

```
cd "H:\Econometrics\project"
```

```
capture log close
```

```
log using "gender_paygap_wage1.log", replace text
```

```
use "http://fmwww.bc.edu/ec-p/data/wooldridge/wage1.dta", clear
```

```
/*
```

```
describing table
```

describe

codebook

misstable summarize

*/

// summarising key variables

summarize wage female educ exper tenure, detail

tab female

// running basic ols regression estimating raw gender wage gap

summarize lwage

reg lwage female

tabstat lwage, by(female)

reg lwage female, vce(robust) // average log wage estimating variance-covariance matrix

// estimates the gender wage gap using heteroskedasticity robust standard

display 100*(exp(_b[female]) - 1)

// gender wage gap controlling for human capital variables

reg lwage female educ exper tenure, vce(robust)

// allow for nonlinear returns to experience

gen exper2 = exper^2

reg lwage female educ exper exper2 tenure, vce(robust)

// Oaxaca decomposition

ssc install oaxaca, replace

```
oaxaca lwage educ exper tenure, by(female) vce(robust)
```

```
tabstat wage lwage educ exper tenure, by(female) stat(mean n)
```

```
// graph to represent wage distribution by gender
```

```
label define gender_label 0 "Male" 1 "Female"
```

```
label values female gender_label
```

```
graph box wage, over(female) ///
```

```
title("Hourly Wage Distribution by Gender")
```

```
graph export "wage_by_gender_boxplot.png", replace
```

```
// histogram of wage distribution
```

```
histogram wage, ///
```

```
title("Distribution of Hourly Wages")
```

```
graph export "wage_histogram.png", replace
```

```
log close
```

```
// estout
```

```
// outreg2
```

Output

name: <unnamed>

log: H:\Econometrics\project\gender_paygap_wage1.log

log type: text

opened on: 21 Mar 2026, 16:32:50

.

. use "http://fmwww.bc.edu/ec-p/data/wooldridge/wage1.dta", clear

.

.

./*

> describing table

> describe

> codebook

> misstable summarize

> */

.

. // summarising key variables

. summarize wage female educ exper tenure, detail

wage

Percentiles		Smallest		
1%	1.67	.53		
5%	2.75	1.43		
10%	2.92	1.5	Obs	526
25%	3.33	1.5	Sum of Wgt.	526
50%	4.65		Mean	5.896103
		Largest	Std. Dev.	3.693086
75%	6.88	21.86		
90%	10	22.2	Variance	13.63888
95%	13	22.86	Skewness	2.007325
99%	20	24.98	Kurtosis	7.970083

female

Percentiles		Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	526
25%	0	0	Sum of Wgt.	526

50%	0		Mean	.4790875
		Largest	Std. Dev.	.500038
75%	1	1		
90%	1	1	Variance	.250038
95%	1	1	Skewness	.0837235
99%	1	1	Kurtosis	1.00701

educ

	Percentiles	Smallest		
1%	4	0		
5%	8	0		
10%	9	2	Obs	526
25%	12	3	Sum of Wgt.	526
50%	12		Mean	12.56274
		Largest	Std. Dev.	2.769022
75%	14	18		
90%	16	18	Variance	7.667485
95%	17	18	Skewness	-.6195741
99%	18	18	Kurtosis	4.884245

exper

	Percentiles	Smallest		
1%	1	1		
5%	1	1		
10%	2	1	Obs	526

25%	5	1	Sum of Wgt.	526
50%	13.5		Mean	17.01711
		Largest	Std. Dev.	13.57216
75%	26	49		
90%	38	49	Variance	184.2035
95%	43	50	Skewness	.7068652
99%	49	51	Kurtosis	2.357318

tenure

	Percentiles	Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	526
25%	0	0	Sum of Wgt.	526
50%	2		Mean	5.104563
		Largest	Std. Dev.	7.224462
75%	7	33		
90%	15	34	Variance	52.19286
95%	22	39	Skewness	2.110273
99%	31	44	Kurtosis	7.658076

. tab female

female	Freq.	Percent	Cum.
-----+-----			

0	274	52.09	52.09
1	252	47.91	100.00

-----+-----

Total	526	100.00
-------	-----	--------

.

```
. // running basic ols regression estimating raw gender wage gap
```

```
. summarize lwage
```

Variable	Obs	Mean	Std. Dev.	Min	Max
----------	-----	------	-----------	-----	-----

-----+-----

lwage	526	1.623268	.5315382	-.6348783	3.218076
-------	-----	----------	----------	-----------	----------

```
. reg lwage female
```

Source	SS	df	MS	Number of obs =	526
--------	----	----	----	-----------------	-----

-----+----- F(1, 524) = 85.04

Model	20.7119996	1	20.7119996	Prob > F =	0.0000
Residual	127.617762	524	.243545348	R-squared =	0.1396

-----+----- Adj R-squared = 0.1380

Total	148.329762	525	.28253288	Root MSE =	.4935
-------	------------	-----	-----------	------------	-------

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
-------	-------	-----------	---	------	----------------------

-----+-----

female	-.3972175	.0430732	-9.22	0.000	-.4818349 - .3126001
_cons	1.81357	.0298136	60.83	0.000	1.755001 1.872139

```
. tabstat lwage, by(female)
```

Summary for variables: lwage

by categories of: female

```
female |    mean
```

```
-----+-----
```

```
0 | 1.81357
```

```
1 | 1.416353
```

```
-----+-----
```

```
Total | 1.623268
```

```
-----
```

```
.
```

```
.
```

```
. reg lwage female, vce(robust) // average log wage estimating variance-covarian
```

```
> ce matrix
```

Linear regression Number of obs = 526

F(1, 524) = 86.36

Prob > F = 0.0000

R-squared = 0.1396

Root MSE = .4935

```
-----
```

```
|       Robust
```

```
lwage |   Coef.   Std. Err.   t   P>|t|   [95% Conf. Interval]
```

```

-----+-----
female | -.3972175 .0427437 -9.29 0.000 -.4811875 -.3132474
_cons | 1.81357 .0323116 56.13 0.000 1.750094 1.877046
-----

```

```

. // estimates the gender wage gap using heteroskedasticity robust standard

```

```

. display 100*(exp(_b[female]) - 1)

```

```

-32.781217

```

```

.

```

```

. // gender wage gap controlling for human capital variables

```

```

. reg lwage female educ exper tenure, vce(robust)

```

```

Linear regression           Number of obs   =    526

```

```

                        F(4, 521)    =    73.76

```

```

                        Prob > F      =    0.0000

```

```

                        R-squared     =    0.3923

```

```

                        Root MSE    =    .41596

```

```

-----+-----
|           Robust
lwage |   Coef.  Std. Err.   t  P>|t|  [95% Conf. Interval]
-----+-----

```

```

female | -.3011459 .0376706 -7.99 0.000  -.3751508  -.2271409

```

```

educ | .0874623 .0079888 10.95 0.000  .0717681  .1031566

```

```

exper | .0046294 .0016195  2.86 0.004  .0014478  .007811

```

```

tenure | .017367 .0035547  4.89 0.000  .0103836  .0243503

```

```

_cons | .5013479 .1152586  4.35 0.000  .2749191  .7277767

```

```

.
. // allow for nonlinear returns to experience
. gen exper2 = exper^2

. reg lwage female educ exper exper2 tenure, vce(robust)

```

```

Linear regression           Number of obs   =    526
                             F(5, 520)      =    75.96
                             Prob > F        =    0.0000
                             R-squared        =    0.4341
                             Root MSE      =    .40179

```

```

-----+-----
      |      Robust
lwage |   Coef.  Std. Err.   t   P>|t|   [95% Conf. Interval]
-----+-----
female | -.2979067   .0365986  -8.14  0.000   -.3698059   -.2260074
educ   |  .0809586   .0076146  10.63  0.000   .0659995   .0959177
exper  |  .03281     .0045721   7.18  0.000   .023828    .041792
exper2 | -.000648    .0000969  -6.69  0.000   -.0008384   -.0004577
tenure |  .016215    .0034715   4.67  0.000   .0093951    .0230348
_cons  |  .4146365    .106826    3.88  0.000   .2047729    .6245002

```

```

.
. // Oaxaca decomposition

```

```
. ssc install oaxaca, replace
checking oaxaca consistency and verifying not already installed...
installing into c:\ado\plus\...
installation complete.
```

```
. oaxaca lwage educ exper tenure, by(female) vce(robust)
```

```
Blinder-Oaxaca decomposition      Number of obs   =    526
                                Model      = linear
Group 1: female = 0                N of obs 1      =    274
Group 2: female = 1                N of obs 2      =    252
```

endowments: $(X1 - X2) * b2$

coefficients: $X2 * (b1 - b2)$

interaction: $(X1 - X2) * (b1 - b2)$

```
-----
|      Robust
lwage |   Coef.  Std. Err.   z   P>|z|   [95% Conf. Interval]
-----+-----
overall |
group_1 |  1.81357  .0324228  55.94  0.000   1.750023   1.877118
group_2 |  1.416353  .0281173  50.37  0.000   1.361244   1.471462
difference | .3972175  .0429164   9.26  0.000   .3131028   .4813321
endowments | .0696264  .0254905   2.73  0.006   .0196659   .1195868
coefficients | .2906308  .0373067   7.79  0.000   .2175111   .3637506
interaction | .0369603  .0223877   1.65  0.099  -.0069188   .0808394
-----+-----
```

```

endowments |
    educ | .0376859 .0202172  1.86  0.062  -.0019391  .0773108
    exper | .0025619 .0035026  0.73  0.465  -.0043031  .0094269
    tenure | .0293786 .0178482  1.65  0.100  -.0056032  .0643605

```

```
-----+-----
```

```

coefficients |
    educ | .1998862 .2096291  0.95  0.340  -.2109792  .6107516
    exper | .0962638 .0546485  1.76  0.078  -.0108453  .2033728
    tenure | .0286981 .0266045  1.08  0.281  -.0234458  .080842
    _cons | -.0342173 .233279  -0.15  0.883  -.4914358  .4230012

```

```
-----+-----
```

```

interaction |
    educ | .0076411 .008903  0.86  0.391  -.0098084  .0250906
    exper | .0066202 .0078891  0.84  0.401  -.0088421  .0220825
    tenure | .022699 .0214851  1.06  0.291  -.019411  .0648089

```

```
-----+-----
```

```
.
```

```
. tabstat wage lwage educ exper tenure, by(female) stat(mean n)
```

Summary statistics: mean, N

by categories of: female

```
female |  wage  lwage  educ  exper  tenure
```

```
-----+-----
```

```
0 | 7.099489  1.81357 12.78832 17.55839 6.474453
```

```
|  274    274    274    274    274
```

```
-----+-----
```

```
1 | 4.587659 1.416353 12.31746 16.42857 3.615079
```

```
| 252 252 252 252 252
```

```
-----+-----
```

```
Total | 5.896103 1.623268 12.56274 17.01711 5.104563
```

```
| 526 526 526 526 526
```

```
-----
```

```
.
```

```
. // graph to represent wage distribution by gender
```

```
. graph box wage, over(female) ///
```

```
> title("Hourly Wage Distribution by Gender")
```

```
. graph export "wage_by_gender_boxplot.png", replace
```

```
(file wage_by_gender_boxplot.png written in PNG format)
```

```
.
```

```
. // histogram of wage distribution
```

```
. histogram wage, ///
```

```
> title("Distribution of Hourly Wages")
```

```
(bin=22, start=.52999997, width=1.1113636)
```

```
. graph export "wage_histogram.png", replace
```

```
(file wage_histogram.png written in PNG format)
```

```
.
```

```
.
```

```
. log close
```

```
name: <unnamed>
```

log: H:\Econometrics\project\gender_paygap_wage1.log

log type: text

closed on: 21 Mar 2026, 16:33:21
